D. Y. PATIL COLLEGE OF ENGINEERING AKURDI, PUNE – 411044

A. Y. 2026 - 27

SUBJECT: 410246: Laboratory Practice III

LAB INDEX **BE Div.: B**

Sr. No	Date of Performance	Title of Experiment	Page No.	Sign/ Remarks
Group A: Design and Analysis of Algorithms				
1		Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity.		
2		Write a program to implement Huffman Encoding using a greedy strategy		
3		Write a program to solve a fractional Knapsack problem using a greedy method.		
4		Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy.		
5		Design n-Queens matrix having first Queen placed. Use backtracking to place remaining Queens to generate the final n-queen's matrix.		
6		**Mini Project** Implement and compare Naive and Rabin-Karp string matching algorithms using the same input.		
Group B: Machine Learning				
1		Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and random forest regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc.		
2		Classify the email using the binary classification method. Email Spam detection has two states: a) Normal State – Not Spam, b) Abnormal State – Spam. Use K-Nearest Neighbors and Support Vector Machine for classification.		
3		Given a bank customer, build a neural network-based classifier that can determine whether they will leave or not in the next 6 months.		
4		Implement Gradient Descent Algorithm to find the local minima of a function.		
5		Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset. Determine the number of clusters using the elbow method.		
6		**Mini Project** Predict Titanic passenger survival using machine learning and passenger data.		
Group C: Blockchain Technology				
1		Installation of MetaMask and study spending Ether per transaction		
2		Create your own wallet using Metamask for crypto transactions.		
3		Write a smart contract on a test network, for Bank account of a customer for following operations: • Deposit money • Withdraw Money • Show balance		

4		Write a program in solidity to create Student data. Use the following constructs: • Structures • Arrays • Fallback Deploy this as smart contract on Ethereum and Observe the transaction fee and Gas values.		
5		Write a survey report on types of Blockchains and its real time use cases.		
6		**Mini Project** Develop a blockchain-based decentralized e-voting application (dApp)		

CERTIFICATE

This is Certified that Mr._____ Roll No: _____of Class BE Computer B has carried out the above practical/termwork with in the four walls of the college. His performance is satisfactory & attendance is _____%

Date: Faculty Member in-charge Head of the Department Principal